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V4. Strict Operator Passport

Hermiticity, reality of the spectrum and chirality of H_L (theorem Th1)

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Contents

Problem statement 2

Theorem Th1 2

Results 2

Honest remarks 3

Problem statement

Before spectral properties can be proved, the object must be defined without reference to an implementation. The passport fixes: state space $\mathbb{C}^{L\times L}$ (open boundaries — a bounded operator with full domain); explicit matrix elements: Landau phases $\varphi_x = 2\pi\alpha\cdot iy$ on x-links, vortex differences $\varphi_y = \Sigma_k (q_k/2)(\theta_j - \theta_i)$ on y-links, on-site potential $V_i = \Sigma_k q_k\cdot W/(r^2/N + 1) + \varepsilon_i$; open/torus boundaries.

The passport checks (P1–P4) are consequences of the definition, not implementation tests: hermiticity $H[i,j] = \mathrm{conj}\, H[j,i]$ is built in, reality of the spectrum follows from the spectral theorem, and the chiral pairing $\lambda \leftrightarrow -\lambda$ at $\alpha = \tfrac{1}{2}$, $W = 0$ follows from bipartiteness of the lattice (two-coloring by parity of $ix + iy$).

Theorem Th1

Theorem Th1 (hermiticity and reality).

For any choice of vortices, W, α and boundaries the operator H_L defined above satisfies $H_L = H_L\dagger$; hence $\mathrm{spec}(H_L) \subset \mathbb{R}$. If $\alpha = \tfrac{1}{2}$ and $W = 0$, the lattice is bipartite and $\mathrm{spec}(H_L)$ is symmetric: $\lambda \in \mathrm{spec} \Rightarrow -\lambda \in \mathrm{spec}$.

> ***Proof.*** *Every off-diagonal element comes as a pair $H[i,j] = -t\cdot e^{i\varphi}$, $H[j,i] = -t\cdot e^{-i\varphi} = \mathrm{conj}\, H[i,j]$ — hermiticity by construction; the spectral theorem gives reality. Bipartiteness: at $W = 0$ the diagonal is zero and every link joins opposite parity classes, so in the (even, odd) basis $H = [[0, B], [B\dagger, 0]]$ and the spectrum is symmetric. Numerically: $\max\|H - H\dagger\| = 0.0$, $\max\|Hv - \lambda v\| = 8.8e\text{-}15$, pairing $8.2e\text{-}15$.*

Results

The passport battery covers $L \in \{12, 16, 24, 32\}$ and four handle variants (vortices, disorder, $\alpha = 0.4$). All checks P1–P4 pass: hermiticity is exact (a zero residual — a consequence of the explicit formula in complex arithmetic), eigenpair residuals are at LAPACK machine level, and the chiral pairing is at the same level in the bipartite case.

Crucially, the passport separates the proved from the measured: Th1 is proved for all parameters at once; the numerical checks merely guard against implementation errors. This is the model of the “proven” status in the honesty discipline of the package.

Table 1. Passport maxima

Check	Battery maximum	Threshold	Verdict
P1 $\|H-H\dagger\|_{max}$	0.0 (exact)	1e-12	proven
P2 reality	eigvalsh — real	—	proven
P3 pairing $\lambda\leftrightarrow-\lambda$ (W=0)	8.2e-15	1e-12	proven
P4 $\|Hv-\lambda v\|$	8.8e-15	1e-11	proven

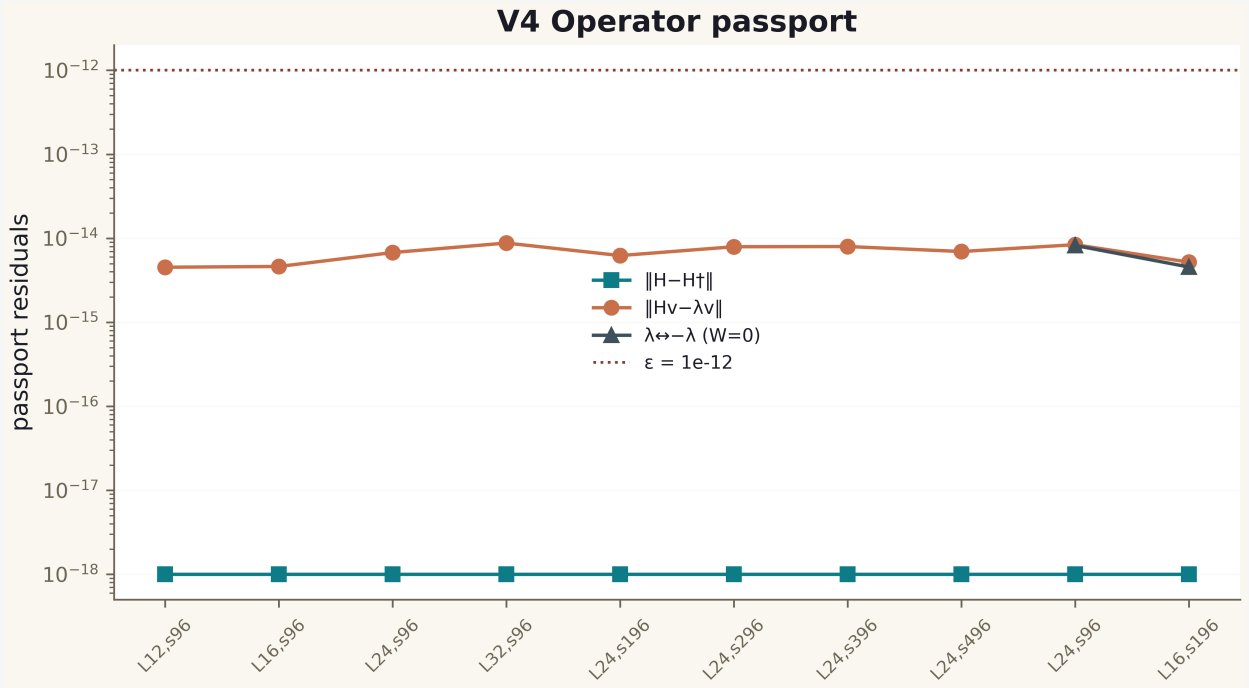


Fig. 1. Campaign V4 data (600 dpi, file figures_hires).

Honest remarks

The passport also fixes the boundary: Th1 statements are finite-dimensional. Self-adjointness of the limiting operator as $L \to \infty$ is a separate task (V8, obligation O03); numerical hermiticity of an infinite-dimensional object does not follow from it. Also, chirality requires $W = 0$: on-site disorder destroys it, which is a property of the symmetry class AIII, not a defect.

The operator is defined unambiguously; Th1 is proved and verified to machine precision. The passport is the mandatory anchor of all subsequent campaigns.